

Practice Questions: Python Basic Data Types

Section: Understanding Data Types

1. What will be the output of the following code?

```
x = 10  
  
print(type(x))
```

2. Which data type will the following expression return?

```
result = 10 / 2
```

3. Write code to declare:

- An integer variable
- A floating-point number
- A complex number

4. Which of the following are valid complex numbers in Python?

- a) 4 + 5j
- b) 3 + j5
- c) 2j
- d) j + 3

5. What is the output of the following?

```
a = 5  
  
b = 2.0  
  
print(type(a + b))
```

Section: String (str) Questions

6. What will be the output of this code?

```
name = "Python"  
  
print(name[1])
```

7. How would you concatenate two strings "Hello" and "World" in Python?

8. What is the result of `len("Data123")`?
9. Write a Python statement to convert a string "123" to an integer.
10. Write a Python statement to convert an integer 456 to a string.

Section: Boolean (bool) Type

11. What is the output of:
- ```
print(bool(0))

print(bool(5))
```
12. What will be the value and data type of the following expression?
- ```
result = 10 > 3
```
13. Which of the following evaluates to False in Python?
- a) `bool([])`
 - b) `bool("Python")`
 - c) `bool(0.0)`
 - d) `bool(1)`

Section: None Type

14. What is the output of this code?
- ```
x = None

print(type(x))
```
15. What is the difference between None and False in Python?
16. Can None be used in comparisons? What will be the output of:
- ```
x = None  
  
print(x == None)
```

Section: Mixed Type Evaluations

17. Predict the output:

```
a = 3 + 2.5
```

```
print(a, type(a))
```

18. Write Python code to check if a variable x is of type int.

19. What will be the result of this expression?

```
x = 10
```

```
y = None
```

```
print(x + y)
```

20. Which function is used to convert:

- A float to an integer?
- A string to a float?
- An integer to a string?